

Additional Topic – Ratios of Double Angles

Key Idea – Ratios of Double Angles

Discussion Questions:

Expressions for $\sin 2A$, $\cos 2A$, $\tan 2A$ may be determined using $2A = A + A$.

(a) Find a formula for $\sin(2A)$ from $\sin(A + B) = \sin A \cos B + \cos A \sin B$

$$\begin{aligned}\sin(2A) &= \sin(A+A) \\ &= \sin A \cos A + \cos A \sin A \\ &= 2 \sin A \cos A\end{aligned}$$

(b) Find a formula for $\cos(2A)$ from $\cos(A + B) = \cos A \cos B - \sin A \sin B$

$$\begin{aligned}\cos(2A) &= \cos(A+A) \\ &= \cos A \cos A - \sin A \sin A \\ &= \cos^2 A - \sin^2 A\end{aligned}$$

(c) Find a formula for $\tan(2A)$ from $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

$$\begin{aligned}\tan(2A) &= \tan(A+A) \\ &= \frac{\tan A + \tan A}{1 - \tan A \tan A} = \frac{2 \tan A}{1 - \tan^2 A}\end{aligned}$$

Summary of Double Angle Formulae

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Discussion Questions:

Consider the identity $\cos 2A = \cos^2 A - \sin^2 A$.

Find two other formulae for $\cos 2A$ by using $\cos^2 A + \sin^2 A = 1$.

$$\begin{aligned}\cos(2A) &= \cos^2 A - \sin^2 A \\ &= (1 - \sin^2 A) - \sin^2 A \\ &= 1 - 2\sin^2 A\end{aligned}$$

$$\begin{aligned}\cos(2A) &= \cos^2 A - \sin^2 A \\ &= \cos^2 A - (1 - \cos^2 A) \\ &= \cos^2 A - 1 + \cos^2 A \\ &= 2\cos^2 A - 1\end{aligned}$$

Therefore

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$\cos 2A = 1 - 2\sin^2 A$$

$$\cos 2A = 2\cos^2 A - 1$$

Summary of Double Angle Formulae

$$\sin 2A = 2\sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$\cos 2A = 1 - 2\sin^2 A$$

$$\cos 2A = 2\cos^2 A - 1$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Exercise:

Find the exact value for the following expressions:

(a) $2\sin 22.5^\circ \cos 22.5^\circ$

using $2\sin A \cos A = \sin(2A)$

$$2\sin 22.5^\circ \cos 22.5^\circ = \sin(2 \times 22.5^\circ) \\ = \sin 45^\circ = \frac{1}{\sqrt{2}}$$

(b) $1 - 2\sin^2 135^\circ$

using $1 - 2\sin^2 A = \cos(2A)$

$$1 - 2\sin^2 135^\circ = \cos(2 \times 135^\circ) \\ = \cos(270^\circ) = 0$$

(c) $\frac{2\sin 75^\circ \cos 75^\circ}{\cos^2 75^\circ - \sin^2 75^\circ} = \frac{\sin(2 \times 75^\circ)}{\cos(2 \times 75^\circ)} = \frac{\sin 150^\circ}{\cos 150^\circ}$

$$= \tan 150^\circ$$
$$= -\tan 30^\circ$$
$$= -\frac{1}{\sqrt{3}}$$

→

- 150° is in Q2
- tan is negative in Q2
- reference angle = $180^\circ - 150^\circ = 30^\circ$

When evaluating or simplifying expressions, it may be necessary to adjust the double angle rules to suit the problem.

For example:

$$\sin 2A = 2 \sin A \cos A$$
$$2 \sin 2A = 4 \sin A \cos A$$
$$\sin 4A = 2 \sin 2A \cos 2A$$

$$\cos 2A = \cos^2 A - \sin^2 A$$
$$2 \cos 2A = 2(\cos^2 A - \sin^2 A)$$
$$\cos 4A = \cos^2 2A - \sin^2 2A$$

Example: Find the exact value for $\sin 15^\circ \cos 15^\circ$.

Step 1: Rearrange the double angle formula to match the format of the question.

$$2 \sin A \cos A = \sin 2A$$
$$\sin A \cos A = \frac{1}{2} \sin 2A$$

Step 2: Substitute

$$\begin{aligned} \sin 15^\circ \cos 15^\circ &= \frac{1}{2} (2 \times 15^\circ) \\ &= \frac{1}{2} \sin 30^\circ \\ &= \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} \end{aligned}$$

Exercise:

Find the exact value for

(a) $2\sin^2 15^\circ - 2\cos^2 15^\circ$.

$$\begin{aligned} &= 2(\sin^2 15^\circ - \cos^2 15^\circ) \\ &= 2\cos(2 \times 15^\circ) \\ &= 2\cos 30^\circ \\ &= 2 \times \frac{\sqrt{3}}{2} \\ &= \sqrt{3} \end{aligned}$$

(b) $6\sin 75^\circ \cos 75^\circ$

$$\begin{aligned} &= 3(2\sin 75^\circ \cos 75^\circ) \\ &= 3\sin(2 \times 75^\circ) \\ &= 3\sin(150^\circ) \\ &= 3 \times \sin 30^\circ \\ &= 3 \times \frac{1}{2} \\ &= \frac{3}{2} \end{aligned}$$

→

- 150 is in Q2
- sin is positive in Q2
- reference angle = $180 - 150 = 30^\circ$

$$\sin 2A = 2\sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$\cos 2A = 1 - 2\sin^2 A$$

$$\cos 2A = 2\cos^2 A - 1$$

$$\tan 2A = \frac{2\tan A}{1 - \tan^2 A}$$

Example:

Find the exact value for $\frac{\sin 2A}{\tan A} - \cos 2A$.

As there is only one identity for $\sin 2A$, substitute this into the expression first then simplify the algebraic fraction.

Then chose the identity for $\cos 2A$, which is most appropriate.

$$\begin{aligned}\frac{\sin 2A}{\tan A} - \cos 2A &= \frac{2 \sin A \cos A}{\tan A} - \cos 2A \\ &= 2 \sin A \cos A \times \frac{\cos A}{\sin A} - \cos 2A \\ &= 2 \cos^2 A - \cos 2A \\ &= 2 \cos^2 A - (2 \cos^2 A - 1) \\ &= 1\end{aligned}$$

$\sin 2A = 2 \sin A \cos A$
$\cos 2A = \cos^2 A - \sin^2 A$
$\cos 2A = 1 - 2 \sin^2 A$
$\cos 2A = 2 \cos^2 A - 1$
$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$

Note: For $\cos 2A$ there are 3 formulae, but the same answer will result whichever formula is chosen. In the above example it is easier to select $\cos 2A = 2 \cos^2 A - 1$

Example:

Prove $(\sin \theta + \cos \theta)^2 \equiv 1 + \sin 2\theta$

$$\begin{aligned} LHS &= (\sin \theta + \cos \theta)^2 \\ &= (\sin^2 \theta + \cos^2 \theta) + (2 \sin \theta \cos \theta) \\ &= 1 + \sin 2\theta \\ &= RHS \end{aligned}$$

$$\begin{aligned} \sin 2A &= 2 \sin A \cos A \\ \cos 2A &= \cos^2 A - \sin^2 A \\ \cos 2A &= 1 - 2 \sin^2 A \\ \cos 2A &= 2 \cos^2 A - 1 \\ \tan 2A &= \frac{2 \tan A}{1 - \tan^2 A} \end{aligned}$$

Exercise:

Prove $\frac{\sin 2A}{1 + \cos 2A} \equiv \tan A$

$$\begin{aligned} LHS &= \frac{\sin 2A}{1 + \cos 2A} \\ &= \frac{2 \sin A \cos A}{1 + (2 \cos^2 A - 1)} \\ &= \frac{2 \sin A \cos A}{1 + 2 \cos^2 A - 1} \end{aligned}$$

$$\begin{aligned} &= \frac{\cancel{2} \sin A \cancel{\cos A}}{\cancel{2} \cos^2 A} \\ &= \frac{\sin A}{\cos A} \\ &= \tan A \\ &= RHS \end{aligned}$$

Key Idea – Solving Equations using Double Angle Identities

Where trigonometric equations have terms with a combination of single and double angles, the double angle identities can be used to reduce the terms to a common angle.

Examples of equations which can be solved using the double angle identities are:

$$\sin 2x - \sin x = 0$$

$$\cos 2x - \cos x = 0$$

$$\sin 2x \sin x = \cos x$$

$$\sin 2x = \cos x$$

Example: Solve the equation $\sin 2x - \cos x = 0$ for $0 \leq x \leq 2\pi$.

$$\sin 2x - \cos x = 0$$

$$2\sin x \cos x - \cos x = 0$$

$$\cos x(2\sin x - 1) = 0$$

$$\text{Therefore } 2\sin x - 1 = 0$$

$$\sin x = \frac{1}{2}$$

$$x = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\text{or } \cos x = 0$$

$$x = 0, \pi, 2\pi$$

$$x = 0, \frac{\pi}{6}, \frac{5\pi}{6}, \pi, 2\pi$$

$$\sin 2A = 2\sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$\cos 2A = 1 - 2\sin^2 A$$

$$\cos 2A = 2\cos^2 A - 1$$

$$\tan 2A = \frac{2\tan A}{1 - \tan^2 A}$$

Exercise:

Solve $\cos 2x + \cos x = 0$ for $0^\circ \leq x \leq 360^\circ$.



$$2\cos^2 x - 1 + \cos x = 0$$

$$2\cos^2 x + \cos x - 1 = 0 \quad (\text{let } u = \cos x)$$

$$2u^2 + u - 1 = 0$$

$$2u^2 + 2u - u - 1 = 0$$

$$2u(u+1) - 1(u+1) = 0$$

$$(u+1)(2u-1) = 0$$



$$u+1 = 0$$

$$u = -1$$

$$\cos x = -1$$

boundary angle

$$x = 180^\circ$$



→ $2u-1=0$

$$2u = 1$$

$$u = \frac{1}{2}$$

$$\cos x = \frac{1}{2}$$

$$x = 60^\circ, 360 - 60^\circ$$

$$= 60^\circ, 300^\circ$$

→ $\cos$ is positive in Q1 & Q4
reference angle = $\cos^{-1}(\frac{1}{2})$
 $= 60^\circ$

$$x = 60^\circ, 180^\circ, 300^\circ$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$\cos 2A = 1 - 2 \sin^2 A$$

$$\cos 2A = 2 \cos^2 A - 1$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Exercise:

Solve $2 - \sin x = 3 \cos 2x$ for $0^\circ \leq x \leq 360^\circ$.

$$2 - \sin x = 3(1 - 2\sin^2 x)$$

$$2 - \sin x = 3 - 6\sin^2 x$$

$$6\sin^2 x - \sin x - 1 = 0 \quad (\text{let } u = \sin x)$$

$$6u^2 - u - 1 = 0$$

$$6u^2 - 3u + 2u - 1 = 0$$

$$3u(2u-1) + 1(2u-1) = 0$$

$$(2u-1)(3u+1) = 0$$

↓

$$2u-1 = 0$$

$$u = \frac{1}{2}$$

$$\sin x = \frac{1}{2}$$

$$x = 30^\circ, 150^\circ$$

$$3u+1 = 0$$

$$u = -\frac{1}{3}$$

$$\sin x = -\frac{1}{3}$$

$$x = 199^\circ 28', 340^\circ 32'$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$\cos 2A = 1 - 2\sin^2 A$$

$$\cos 2A = 2\cos^2 A - 1$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Class Questions:

Ratios of Double Angles

1. Use a double angle identity to find the exact value of the following:

(i) $\sin 90^\circ$ (ii) $\cos 120^\circ$ (iii) $\tan 60^\circ$

2. Find exact values for $\sin 2A$, $\tan 2A$ and $\cos 2A$ if $0 < A < \frac{\pi}{2}$:

(i) $\sin A = \frac{8}{17}$ (ii) $\cos A = \frac{12}{13}$ (iii) $\tan A = \frac{15}{8}$

3. Find exact values for the following using the double angle rules:

(i) $2\sin 15^\circ \cos 15^\circ$ (ii) $\sin 22.5^\circ \cos 22.5^\circ$

(iii) $\frac{2\sin 75^\circ \cos 75^\circ}{\cos^2 75^\circ - \sin^2 75^\circ}$ (iv) $2\sin 67.5^\circ \cos 67.5^\circ$

(v) $\cos^2 \frac{\pi}{12} - \sin^2 \frac{\pi}{12}$

4. Simplify:

(i) $\frac{\sin 2x}{1 + \cos 2x}$ (ii) $\frac{\sin 2A}{\tan A} - \cos 2A$

5. Prove the following:

(i) $(\sin \theta + \cos \theta)^2 \equiv 1 + \sin 2\theta$ (ii) $\frac{\sin 2\theta}{1 + \cos 2\theta} \equiv \tan \theta$

(iii) $\frac{1 - \cos 2A}{\sin 2A} \equiv \tan A$

6. Solve the following equations

(i) $\sin 2x = 3\cos x \quad 0 \leq x \leq 2\pi$ (ii) $\cos 2x = \cos x \quad 0^\circ \leq x \leq 360^\circ$

Solutions to Class Questions:

Ratios of Double Angles

1.

(i)

$$\begin{aligned}\sin(90^\circ) &= \cos(2 \times 45^\circ) \\ &= 2 \sin(45^\circ) \cos(45^\circ) \\ &= 2 \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \\ &= 2 \times \frac{1}{2} \\ &= 1\end{aligned}$$

(ii)

$$\begin{aligned}\cos(120^\circ) &= \cos(2 \times 60^\circ) \\ &= \cos^2(60^\circ) - \sin^2(60^\circ) \\ &= \left(\frac{1}{2}\right)^2 - \left(\frac{\sqrt{3}}{2}\right)^2 \\ &= \frac{1}{4} - \frac{3}{4} \\ &= -\frac{1}{2}\end{aligned}$$

(iii)

$$\begin{aligned}\tan(60^\circ) &= \tan(2 \times 30^\circ) \\ &= \frac{2 \tan 30^\circ}{1 - \tan^2 30^\circ} \\ &= \frac{2 \times \frac{1}{\sqrt{3}}}{1 - \left(\frac{1}{\sqrt{3}}\right)^2} \\ &= \frac{2 \times \frac{1}{\sqrt{3}}}{1 - \frac{1}{3}} \\ &= \frac{\frac{2}{\sqrt{3}}}{\frac{2}{3}} = \frac{2}{\sqrt{3}} \times \frac{3}{2} \\ &= \frac{3}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \\ &= \sqrt{3}\end{aligned}$$

2. (i) If $\sin A = \frac{8}{17}$ then $\cos A = \frac{15}{17}$

and $\tan A = \frac{8}{15}$

$$\begin{aligned}\sin 2A &= 2 \sin A \cos A \\ &= 2 \left(\frac{8}{17}\right) \left(\frac{15}{17}\right) \\ &= \frac{240}{289}\end{aligned}$$

$$\begin{aligned}\cos 2A &= 2 \cos^2 A - 1 \\ &= 2 \left(\frac{15}{17}\right)^2 - 1 \\ &= \frac{161}{289}\end{aligned}$$

$$\begin{aligned}\tan 2A &= \frac{2 \tan A}{1 - \tan^2 A} \\ &= \frac{2 \left(\frac{8}{15}\right)}{1 - \left(\frac{8}{15}\right)^2} \\ &= \frac{240}{161}\end{aligned}$$

(iii) If $\tan A = \frac{15}{8}$ then $\sin A = \frac{15}{17}$ and $\cos A = \frac{8}{17}$

$$\begin{aligned}\sin 2A &= 2 \sin A \cos A \\ &= 2 \left(\frac{15}{17}\right) \left(\frac{8}{17}\right) \\ &= \frac{240}{289}\end{aligned}$$

$$\begin{aligned}\cos 2A &= 2 \cos^2 A - 1 \\ &= 2 \left(\frac{8}{17}\right)^2 - 1 \\ &= -\frac{161}{289}\end{aligned}$$

(ii) If $\cos A = \frac{12}{13}$ then $\sin A = \frac{5}{13}$

and $\tan A = \frac{5}{12}$

$$\begin{aligned}\sin 2A &= 2 \sin A \cos A \\ &= 2 \left(\frac{5}{13}\right) \left(\frac{12}{13}\right) \\ &= \frac{120}{169}\end{aligned}$$

$$\begin{aligned}\cos 2A &= 2 \cos^2 A - 1 \\ &= 2 \left(\frac{12}{13}\right)^2 - 1 \\ &= \frac{119}{169}\end{aligned}$$

$$\begin{aligned}\tan 2A &= \frac{2 \tan A}{1 - \tan^2 A} \\ &= \frac{2 \left(\frac{5}{12}\right)}{1 - \left(\frac{5}{12}\right)^2} \\ &= \frac{120}{119}\end{aligned}$$

$$\begin{aligned}\tan 2A &= \frac{2 \tan A}{1 - \tan^2 A} \\ &= \frac{2 \left(\frac{15}{8}\right)}{1 - \left(\frac{15}{8}\right)^2} \\ &= -\frac{240}{161}\end{aligned}$$

3. (i) $2 \sin 15^\circ \cos 15^\circ = \sin 30^\circ$

$$= \frac{1}{2}$$

(ii)

$$\begin{aligned} \sin 22.5^\circ \cos 22.5^\circ &= \frac{1}{2} \sin 45^\circ \\ &= \frac{1}{2} \times \frac{1}{\sqrt{2}} \\ &= \frac{1}{2\sqrt{2}} \\ &= \frac{\sqrt{2}}{4} \end{aligned}$$

(iv)

$$\begin{aligned} 2 \sin \theta \cos \theta &= \sin 2\theta \\ 2 \sin 67.5^\circ \cos 67.5^\circ &= \sin(2 \times 67.5^\circ) \\ &= \sin 135^\circ \\ &= \sin 45^\circ \\ &= \frac{1}{\sqrt{2}} \end{aligned}$$

4.

(i)

$$\begin{aligned} \frac{\sin 2x}{1 + \cos 2x} &= \frac{2 \sin x \cos x}{1 + 2 \cos^2 x - 1} \\ &= \frac{2 \sin x \cos x}{2 \cos^2 x} \\ &= \frac{\sin x}{\cos x} \\ &= \tan x \end{aligned}$$

(iii)

$$\begin{aligned} \frac{2 \sin 75^\circ \cos 75^\circ}{\cos^2 75^\circ - \sin^2 75^\circ} &= \frac{\sin 150^\circ}{1 - \sin^2 75^\circ - \sin^2 75^\circ} \\ &= \frac{\sin 150^\circ}{1 - 2 \sin^2 75^\circ} \\ &= \frac{\sin 150^\circ}{\cos 150^\circ} \\ &= \tan 150^\circ \\ &= -\frac{1}{\sqrt{3}} = -\frac{\sqrt{3}}{3} \end{aligned}$$

(v)

$$\begin{aligned} \cos^2 \theta - \sin^2 \theta &= \cos 2\theta \\ \cos^2 \frac{\pi}{12} - \sin^2 \frac{\pi}{12} &= \cos \left(2 \times \frac{\pi}{12} \right) \\ &= \cos \left(\frac{\pi}{6} \right) \\ &= \frac{\sqrt{3}}{2} \end{aligned}$$

(ii)

$$\begin{aligned} \frac{\sin 2A}{\tan A} - \cos 2A &= \frac{2 \sin A \cos A}{\frac{\sin A}{\cos A}} - (2 \cos^2 A - 1) \\ &= \frac{2 \sin A \cos A \cos A}{\sin A} - (2 \cos^2 A - 1) \\ &= 2 \cos^2 A - 2 \cos^2 A + 1 \\ &= 1 \end{aligned}$$

5. (i) $(\sin \theta + \cos \theta)^2 = 1 + \sin 2\theta$

$$\begin{aligned} LHS &= (\sin \theta + \cos \theta)^2 \\ &= \sin^2 \theta + 2 \sin \theta \cos \theta + \cos^2 \theta \\ &= \sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta \\ &= 1 + \sin 2\theta \\ &= RHS \end{aligned}$$

(ii)

$$\begin{aligned} \frac{\sin 2\theta}{1 + \cos 2\theta} &\equiv \tan \theta \\ LHS &= \frac{2 \sin \theta \cos \theta}{1 + (1 - 2 \sin^2 \theta)} \\ &= \frac{2 \sin \theta \cos \theta}{2 \cos^2 \theta} \\ &= \frac{\sin \theta}{\cos \theta} \\ &= \tan \theta \\ &= RHS \end{aligned}$$

(iii)

$$\begin{aligned} \frac{1 - \cos 2A}{\sin 2A} &\equiv \tan A \\ LHS &= \frac{1 - \cos 2A}{\sin 2A} \\ &= \frac{1 - (1 - 2 \sin^2 A)}{2 \sin A \cos A} \\ &= \frac{2 \sin^2 A}{2 \sin A \cos A} \\ &= \frac{\sin A}{\cos A} \\ &= \tan A \end{aligned}$$

6. (i) $\sin 2x = 3 \cos x \quad 0 \leq x \leq 2\pi$
 $2 \sin x \cos x - 3 \cos x = 0$

$$\cos x (2 \sin x - 3) = 0$$

$$\cos x = 0$$

$$x = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$2 \sin x = 3 \quad \text{Not Possible}$$

$$\sin x = 1.5$$

(ii) $\cos 2x = \cos x \quad 0^\circ \leq x \leq 360^\circ$
 $\cos 2x = \cos x$

$$2 \cos^2 x - 1 = \cos x$$

$$2 \cos^2 x - \cos x - 1 = 0$$

$$\text{Let } u = \cos x \quad 2u^2 - u - 1 = 0$$

$$(2u + 1)(u - 1) = 0$$

$$2u + 1 = 0$$

$$u = -0.5$$

$$\cos x = -0.5$$

$$x = \cos^{-1} 0.5$$

$$= 180^\circ - 60^\circ, 180^\circ + 60^\circ$$

$$= 120^\circ, 240^\circ$$

$$u - 1 = 0$$

$$u = 1$$

$$\cos x = 1$$

$$x = \cos^{-1} 1$$

$$= 0^\circ, 360^\circ$$